

Joey Ma  
September 25th, 2026

## CSCI596 Assignment 3—Parallel Computation of $\pi$ and Scalability Analysis

global\_pi.c, global\_pi\_iso.c, and global\_pi.out files are attached as part of this submission. The content of global\_pi.c was also pasted below as well.

### 1. Programming

#### 1. *Submit the source code of global\_pi.c.*

```
#include "mpi.h"
#include <stdio.h>
#define NBIN 10000000000
/* #define NPERP 10000000000 */

int nprocs; /* Number of processes */
int myid; /* My rank */

double global_sum(double partial) {
    /* Write your hypercube algorithm here */
    double hisdone;
    MPI_Status status;
    double mydone = partial;
    for (int bitvalue=1; bitvalue<nprocs; bitvalue*=2)
    {
        int partner = myid ^ bitvalue;
        //send mydone to partner;
        MPI_Send(&mydone, 1, MPI_DOUBLE, partner, bitvalue, MPI_COMM_WORLD);
        //receive hisdone from partner;
        MPI_Recv(&hisdone, 1, MPI_DOUBLE, partner, bitvalue, MPI_COMM_WORLD,
&status);
        mydone = mydone + hisdone;
    }
    return mydone;
}

int main(int argc, char *argv[]) {
    //double partial, sum, avg;
    double partial;
    double cpu1, cpu2;
    long long i;
    double step, x, sum=0.0, pi;
    //long long NBIN;
```

```

MPI_Init(&argc, &argv);
MPI_Comm_rank(MPI_COMM_WORLD, &myid);
MPI_Comm_size(MPI_COMM_WORLD, &nprocs);

//partial = (double) myid;
//printf("Node %d has partial value %le\n", myid, partial);

cpu1 = MPI_Wtime();
//NBIN = (long long)NPERP * nprocs;
    step = 1.0/NBIN;
    //for (i=0; i<NBIN; i++) {
    for (i=myid; i<NBIN; i+=nprocs) {
        x = (i+0.5)*step;
        sum += 4.0/(1.0+x*x);
    }
    //pi = sum*step;
    partial = sum*step;
//sum = global_sum(partial);
pi = global_sum(partial);
cpu2 = MPI_Wtime();

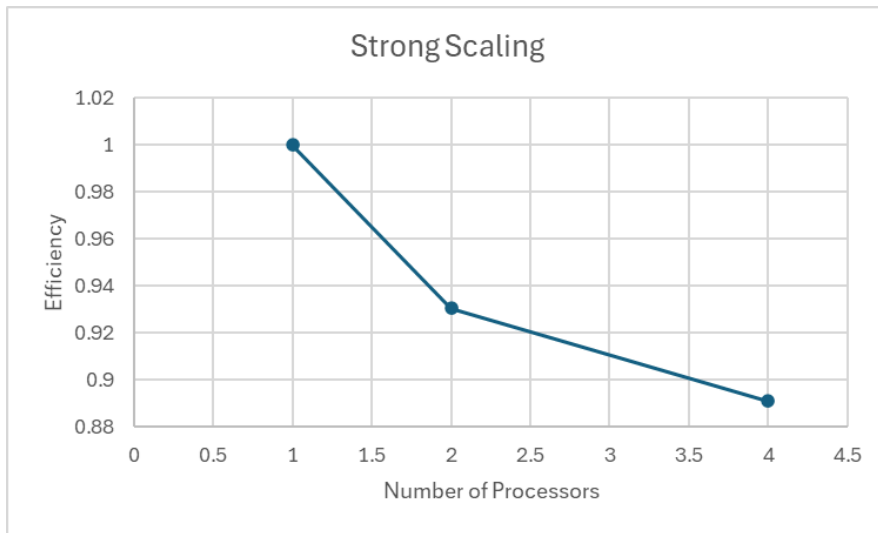
if (myid == 0) {
    //avg = sum/nprocs;
    //printf("Global average = %le\n", avg);
    printf("Pi = %le\n", pi);
    //printf("Execution time (s) = %le\n", cpu2-cpu1);
    printf("Nprocs & Execution time (s) = %d\t%le\n", nprocs, cpu2-cpu1);
}

MPI_Finalize();
return 0;
}

```

## 2. Scalability

2. (Fixed problem-size scaling). Submit the plot.



3. (Isogranular scaling). Submit the plot.

